



CIS 4951 Capstone II | Giggso

# Team A: Critical Materials Recovery

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Decision support for identifying valuable or critical materials in cataloged items.

## TEAM MEMBERS

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# Problem Understanding: Fragmented Data & Inefficient Material Assessment



## OPERATIONAL CHALLENGE

### Core Operational Gap

Organizations receiving surplus, obsolete, damaged, or unserviceable items lack an efficient method to determine if those items contain critical, strategic, precious, or valuable materials.

## INFORMATION BOTTLENECK

### Distributed Data Sources

Currently, relevant information is distributed across:

- Public item catalogs
- Technical descriptions
- Material references
- Lifecycle records
- Commodity publications

**This fragmentation makes it difficult to quickly and accurately assess material composition.**

# Why This Matters: Operational & Compliance Risks of Manual Research



## CORE CHALLENGE

### Manual Research Bottlenecks

Manual research into material composition is slow and produces inconsistent results. This creates significant operational and compliance risks.

Relying on ungrounded manual processes or unsupported AI inferences without a traceable source introduces severe operational exposure across the organization.

#### Missed Recovery & Retention Opportunities

Missed opportunities to retain useful items, recover scarce materials, or recycle valuable content.

#### Inconsistent Analytical Results

Slow, manual evaluations yield variable and non-standardized material assessments.

#### Poor Recommendation Documentation

Manual processes consistently fail to properly document why a specific recommendation was made.

#### Lack of Verifiable Proof & Auditability

Inability to provide traceable audit trails or objective proof for compliance checks.

#### Risk of Unsupported AI Assumptions

Relying on ungrounded AI inferences without a traceable source introduces severe operational risk.

## EXPECTED OUTCOME

# Traceable Decision-Support System Prototype

Building a reproducible prototype to automate item matching, material analysis, and disposition recommendations with full provenance.

## CORE SYSTEM MANDATE

Every system recommendation must explicitly expose underlying evidence, decision rules, uncertainty metrics, and source provenance.

## PROTOTYPE ARCHITECTURE & PIPELINE FLOW



### 01 / INGESTION

#### Catalog Matching

Accepts a public item identifier or description and automatically retrieves matching public catalog information.

### 02 / ANALYSIS

#### Critical Mineral Mapping

Connects item data to material information and compares components with current critical-material references.

### 03 / DECISION ENGINE

#### Action Recommendations

Recommends specific lifecycle paths:

- Retention or Reclamation
- Recycling or Manual Review
- Ordinary Disposal

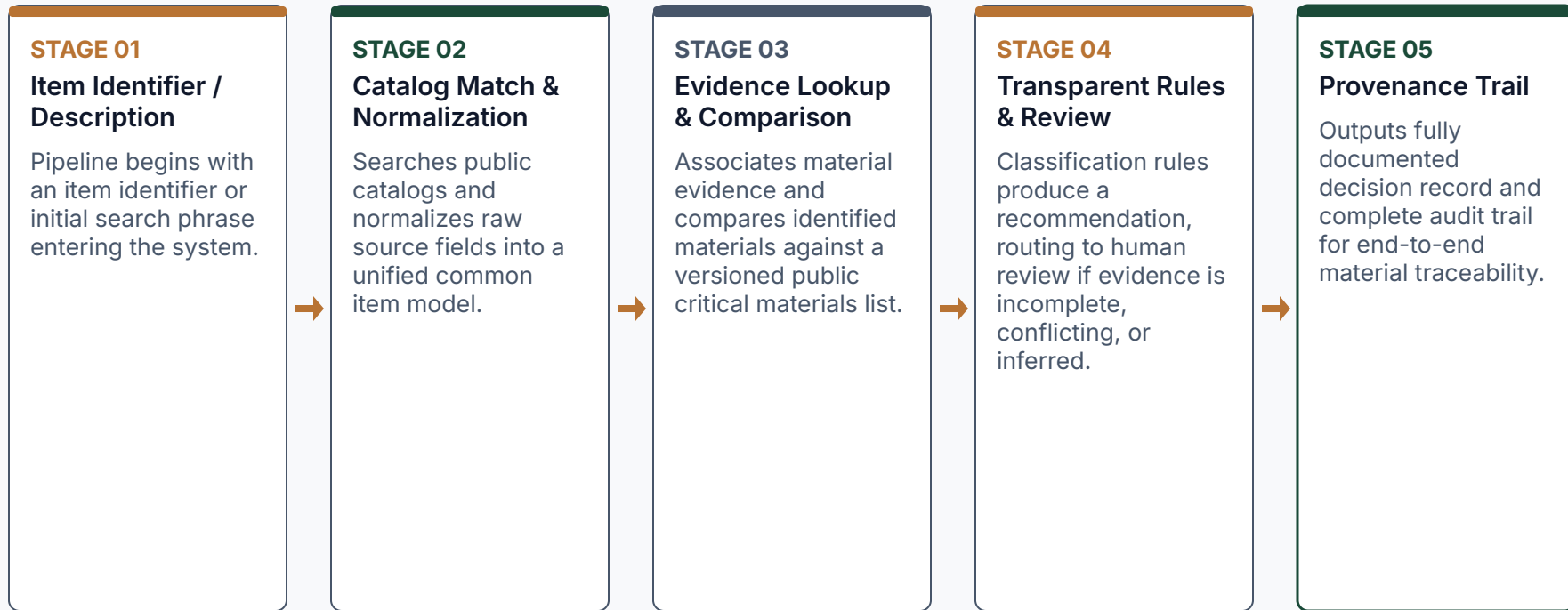
### 04 / TRACEABILITY

#### Evidence & Provenance

Every recommendation explicitly presents supporting evidence, underlying rules, uncertainty metrics, and source provenance.

# Core Workflow

## End-to-End Decision & Provenance Pipeline



# Key Data Entities: Core Structural & Provenance Graph Model



## Category 01- Item Structure

### Item:

- Physical asset or cataloged item being evaluated for potential material recovery.

### Component:

- Part or sub-assembly of an item that may contain identifiable materials.

### Material:

- Substance identified or potentially present within an item or component.

## Category 02- Evidence & References

### Composition Evidence:

- Source information supporting what materials an item or component contains.

### Criticality Reference:

- Reference used to determine whether an identified material is considered critical, strategic, precious, or otherwise important.

## Category 03- Recovery & Decision Report

### Recovery Method:

- Potential method for reclaiming, recycling, retaining, or otherwise handling an identified material.

### Value Estimate:

- Preliminary estimate of potential recovery or economic value, where sufficient data exists.

### Recommendation:

- System-generated suggested action based on available evidence and defined rules.

## Category 04- Traceability & Review

### Human Review Decision

Reviewer confirmation, revision, or rejection of a system recommendation.

### Provenance Record

Traceable record showing the sources, evidence, rules, and processing steps behind a recommendation.

### Audit Event

Recorded action or change associated with an analysis or review.

# Initial Data Sources & References



## PUBLIC LOGISTICS

### PUB LOG

Identified as a potential source for non-restricted item identifiers, baseline metadata, and logistics characteristics.

## MINERAL REFERENCES

### USGS Geological Datasets

- **USGS Critical Minerals List**  
Potential reference for determining whether identified materials are currently classified as critical.
- **USGS Mineral Commodity Summaries**  
Provides additional mineral, supply, and commodity context.

## MATERIALS & SYNTHETIC DATA

**NIST Materials Data Repository** — Potential standardized materials information.

**EPA RCRAInfo** — Optional environmental/compliance reference.

**Synthetic/Test Records** — Testing and validation when verified public examples are limited.

## DATA ACCESS & LICENSING

Public availability does not automatically mean open licensing. Source permissions and usage restrictions will be verified before integration

# Proposed Technical Approach

## Data Normalization & Rule-Based Inference



### STAGE 01 — DATA MODELING

#### Catalog Data Normalization

Initial technical build begins by normalizing raw source fields from disparate supplier and internal catalogs into a unified, common item model.

### STAGE 02 — MATERIAL VERIFICATION

#### Material Identification & Verification

Identified materials will be compared against relevant public references to determine potential criticality, strategic importance, or recovery value.

### Transparent Recommendation Rules

#### Proposed Recommendation Rules

Explore clear, traceable rules for generating potential recovery, recycling, retention, or review recommendations based on available evidence.

### Proposed Human Review Triggers

#### EXPLICIT OVERRIDE CONDITIONS:

- Material composition is inferred rather than explicit
- Evidence or underlying catalog data is incomplete
- Source datasets or catalog records conflict
- A potentially hazardous material is identified

### Traceability & Audit Trail

Complete, permanent logging of all raw inputs, source records, and active rules applied for full system traceability.

# Human Review & Explainability

## Transparent Evidence Display & Decision Framework



### 01 | ITEM IDENTIFICATION

#### Matched Item Record

Display the catalog item and source information used to identify it.

### 02 | COMPOSITION SPECTRUM

#### Identified / Potential Materials

Show the materials supported by available catalog information and external evidence.

### 03 | SYSTEM ACCURACY

#### Evidence Confidence

Indicate the strength and completeness of the evidence supporting the identification.

### 04 | PROVENANCE & LOGIC TRACE

#### Supporting Evidence & Classification Rules

Show the sources and decision rules that led to the recommendation. Clearly flag missing, inferred, or conflicting information.

### 05 | DECISION CONTROL & AUDIT

#### Human Action & Review Triggers

##### Confirmation When Needed

Route uncertain or conflicting cases to a reviewer who can:

- **ACCEPT** — Confirm recommendation
- **REVISE** — Correct information
- **REJECT** — Flag recommendation as unsupported

# Proposed Evaluation Plan: Initial Metrics & Validation Approach



## 100+

Public or Synthetic Item Records

Benchmark Dataset Target

## Precision / Recall

Material Identification Rate

Core Accuracy Metric

### Provenance Completeness:

Measure whether each recommendation can be traced back to its supporting sources, evidence, and applied rules.

### Data

- **Benchmark Dataset:** Develop a set of public and/or synthetic item records with documented expected outcomes for testing.
- **Catalog-Match Success:** Quantitative rate measuring exact matches against cataloged items.
- **Material-ID Performance:** Use precision and recall to measure how accurately the system identifies relevant materials.

### Human Review & Operational Efficiency

- **Uncertain Case Routing:** Percentage of low-confidence or ambiguous cases correctly routed to human reviewers.
- **Reviewer Agreement & Speed:** Tracking inter-reviewer agreement rate alongside overall average processing times.
- **Data Limitation Documentation:** Explicitly logging known edge cases, missing data, and system constraints.

# Architecture & Tooling: Roles of Aryx and Raven



## DATA & CONTEXT LAYER

### Aryx

Serves as the candidate context and data layer supporting key architectural representations:

- Knowledge graph representation
- Provenance tracking
- Entity relationships required for item-material mapping

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## DEV WORKFLOW GUARDRAIL

### Raven

Acts as a development workflow guardrail maintaining engineering discipline:

- Reducing AI comprehension debt
- Executing security checks
- Maintaining documentation discipline

## SCOPE BOUNDARY & SYSTEM INTEGRITY

**Neither tool replaces** the project-specific recommendation logic required for material classification.



# Sprint Plan: Delivering an End-to-End Thin Slice

## STEP 01

### Alignment & Item Selection

- **PO Alignment:** Confirm the smallest acceptable demo with the Product Owner.
- **Sample Selection:** Select 5–10 public or synthetic item examples for initial validation.
- **Scope Boundaries:** Establish baseline criteria for the sprint demo artifact.



## STEP 02

### Schema & Source Setup

- **Initial Schema:** Define entities for Items, Materials, Evidence, Criticality References, Recommendations, and Provenance.
- **Source Integration:** Configure one primary public source alongside a synthetic fallback.



## STEP 03

### Execution & Evaluation

- **Thin-Slice Build:** Deliver one end-to-end evidence-backed recommendation sample.
- **Substrate Eval:** Evaluate Aryx as our core graph substrate.
- **Backlog Readiness:** Complete Odoo ticket preparation for ongoing workstreams.

# Team Contribution & Workstream Allocation



## Scoping & Research

- Requirements scoping
- Data source research



## Schema & Provenance Architecture

- Schema design
- Provenance design



## Implementation & Evaluation

- Prototype implementation
- Developing evaluation plan
- Creating presentations
- Tracking Odoo tickets

### GROUP COORDINATION & COMMUNICATION

Group coordination is maintained by sharing daily project hours and working schedules to ensure consistent progress and communication.



# Initial Questions for Product Owner (Giggso)

## CATEGORY 01

### Data Priorities

#### Catalog Source Hierarchy

Confirm the explicit priority order for integration across catalog sources:

1. PUB LOG
2. USGS
3. NIST

## CATEGORY 02

### System Scope

#### Sprint 1 Demo Size

Define the smallest acceptable baseline scope for the initial Sprint 1 demonstration.

#### Recovery Value Estimates

Clarify whether illustrative recovery value estimates should be included in the early project phase.

## CATEGORY 03

### Governance

#### Evidence Quality

Establish a standardized evidence-quality labeling scheme across data entities.

#### Human Reviewer Persona

Define the intended persona and workflow expectations for the human reviewer.

#### Repository Boundaries

Establish public repository data boundaries and security guardrails.

## FINAL SUMMARY

# Conclusion

Traceable, Evidence-Backed  
Decision Support

## PROJECT TARGET

Zero Unsupported AI  
Assumptions

*"Team A's mission is to deliver a traceable, reproducible decision-support prototype connecting item records to verifiable material evidence."*

### - MISSION STATEMENT

The system will not treat an AI-generated inference as a confirmed material composition unless it is supported by a traceable source.

By relying on transparent rules and explicit human review thresholds, we will mitigate the risks of unsupported AI guesses and manual research inefficiencies.

